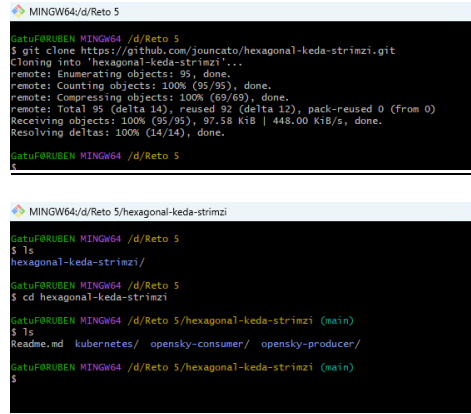


Reto 5

1.-Captura de pantalla del comando git clone y el árbol de carpetas mostrado con ls o tree:

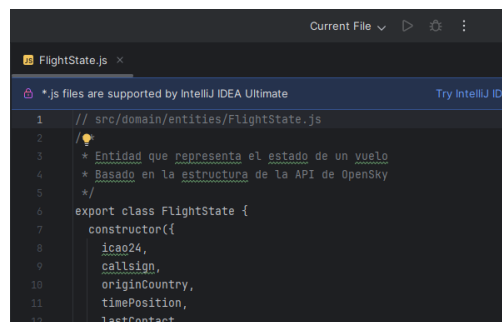


```
MINGW64/d/Reto 5
$ git clone https://github.com/jouncato/hexagonal-keda-stimzi.git
Cloning into 'hexagonal-keda-stimzi'...
remote: Enumerating objects: 95, done.
remote: Counting objects: 100% (95/95), done.
remote: Compressing objects: 100% (69/69), done.
remote: Total 95 (delta 14), reused 92 (delta 12), pack-reused 0 (from 0)
Receiving objects: 100% (95/95), 97.58 KiB | 448.00 KiB/s, done.
Resolving deltas: 100% (14/14), done.

MINGW64/d/Reto 5/hexagonal-keda-stimzi
$ ls
hexagonal-keda-stimzi/
$ cd hexagonal-keda-stimzi
$ ls
Readme.md  kubernetes/  opensky-consumer/  opensky-producer/
$
```

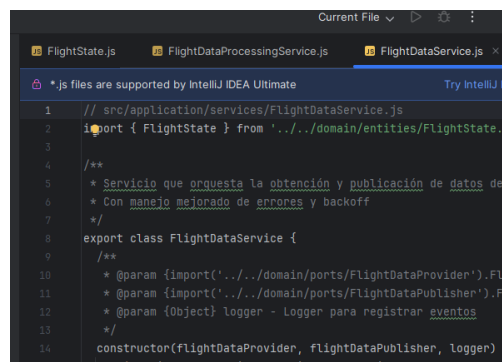
2.-Fragmento de código:

* Domain:



```
// src/domain/entities/FlightState.js
/**
 * Entidad que representa el estado de un vuelo
 * Basado en la estructura de la API de OpenSky
 */
export class FlightState {
  constructor({
    icao24,
    callsign,
    originCountry,
    timePosition,
    lastContact
  }) {}
}
```

* Application:



```
// src/application/services/FlightDataService.js
import { FlightState } from '../../domain/entities/FlightState.js'

/**
 * Servicio que orquesta la obtención y publicación de datos de vuelo
 * Con manejo mejorado de errores y backoff
 */
export class FlightDataService {
  /**
   * @param {import('../../domain/ports/FlightDataProvider').FlightDataProvider} provider
   * @param {import('../../domain/ports/FlightDataPublisher').FlightDataPublisher} publisher
   * @param {Object} logger - Logger para registrar eventos
   */
  constructor(flightDataProvider, flightDataPublisher, logger) {
    this.flightDataProvider = flightDataProvider;
  }
}
```

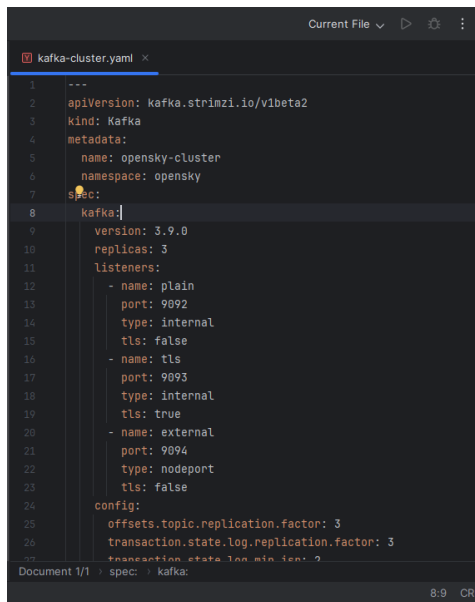
* Infrastructure:

```
Current File ▾ ▶ ⚙ ⋮ 👤 🔍 ⚙
ConsoleRepositoryAdapter.js KafkaProducerAdapter.js x OpenSkyApiAdapter.js
*.js files are supported by IntelliJ IDEA Ultimate Try IntelliJ IDEA Ultimate Learn more
1 // src/infrastructure/adapters/KafkaProducerAdapter.js
2 import { Kafka, CompressionTypes, Partitioners } from 'kafkajs';
3 import { FlightDataPublisher } from '../../domain/ports/FlightDataPublisher.js';
4 import { kafkaConfig } from '../../config/kafka.js';
5
6 /**
7  * Adaptador para Kafka Producer
8  * Implementa el puerto FlightDataPublisher
9  */
10 export class KafkaProducerAdapter extends FlightDataPublisher {
11   /**
12    * @param {Object} config - Configuración para Kafka
13    * @param {Object} logger - Logger para registrar eventos
14    */
15   constructor(config = kafkaConfig, logger) {
16     super();
17     this.config = config;
18     this.logger = logger;
19   }
20 }
```

```
Current File ▾ ▶ ⚙ ⋮ 👤 🔍 ⚙
KafkaProducerAdapter.js OpenSkyApiAdapter.js x
*.js files are supported by IntelliJ IDEA Ultimate Try IntelliJ IDEA Ultimate Learn more
1 // src/infrastructure/adapters/OpenSkyApiAdapter.js
2 import axios from 'axios';
3 import { FlightDataProvider } from '../../domain/ports/FlightDataProvider.js';
4 import { openSkyConfig } from '../../config/api.js';
5
6 /**
7  * Adaptador para la API de OpenSky con manejo mejorado de límites de tasa
8  * Implementa el puerto FlightDataProvider
9  */
10 export class OpenSkyApiAdapter extends FlightDataProvider {
11   /**
12    * @param {Object} config - Configuración para la API
13    * @param {Object} logger - Logger para registrar eventos
14    */
15   constructor(config = openSkyConfig, logger) {
16     super();
17     this.config = config;
18     this.logger = logger;
19     this.lastRequestTime = 0;
20   }
21 }
```

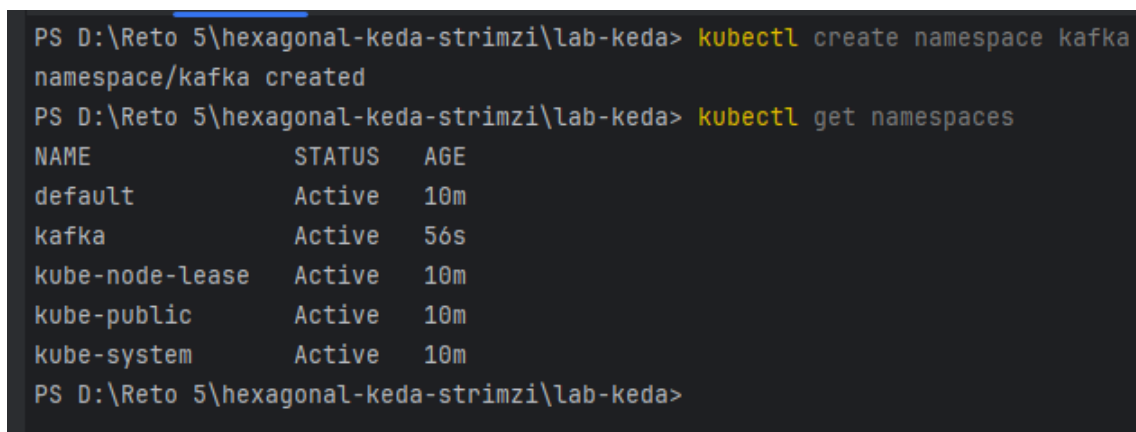
```
Current File ▾ ▶ ⚙ ⋮ 👤 🔍 ⚙
KafkaProducerAdapter.js OpenSkyApiAdapter.js HealthServer.js x ConsoleInterface.js
*.js files are supported by IntelliJ IDEA Ultimate Try IntelliJ IDEA Ultimate Learn more
1 // src/infrastructure/adapters/HealthService.js
2 import http from 'http';
3 import fs from 'fs';
4 import path from 'path';
5
6 /**
7  * Servicio de salud para monitoreo de la aplicación
8  */
9 export class HealthService {
10   /**
11    * @param {Object} logger - Logger para registrar eventos
12    * @param {number} port - Puerto para el servidor HTTP
13    */
14   constructor(logger, port = 8080) {
15     this.logger = logger;
16     this.port = port;
17     this.server = null;
18     this.isHealthy = true;
19     this.statusFile = process.env.HEALTH_STATUS_FILE || './app/health/status';
20   }
21 }
```

3.-Estructura YAML del deployment de Strimzi:



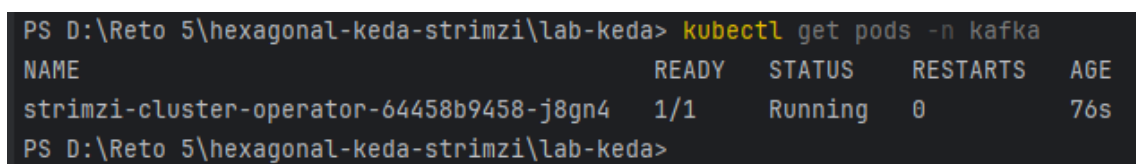
```
1 ---
2 apiVersion: kafka.strimzi.io/v1beta2
3 kind: Kafka
4 metadata:
5   name: opensky-cluster
6   namespace: opensky
7 spec:
8   kafka:
9     version: 3.9.0
10    replicas: 3
11    listeners:
12      - name: plain
13        port: 9092
14        type: internal
15        tls: false
16      - name: tls
17        port: 9093
18        type: internal
19        tls: true
20      - name: external
21        port: 9094
22        type: nodeport
23        tls: false
24    config:
25      offsets.topic.replication.factor: 3
26      transaction.state.log.replication.factor: 3
27      transaction.state.log.min.isr: 2
```

4.- Screenshot del comando de creación del namespace y del despliegue de pods en el estado "Running":



```
PS D:\Reto 5\hexagonal-keda-strimzi\lab-keda> kubectl create namespace kafka
namespace/kafka created
PS D:\Reto 5\hexagonal-keda-strimzi\lab-keda> kubectl get namespaces

NAME                STATUS    AGE
default              Active    10m
kafka                 Active    56s
kube-node-lease      Active    10m
kube-public          Active    10m
kube-system          Active    10m
PS D:\Reto 5\hexagonal-keda-strimzi\lab-keda>
```



```
PS D:\Reto 5\hexagonal-keda-strimzi\lab-keda> kubectl get pods -n kafka

NAME                                     READY   STATUS    RESTARTS   AGE
strimzi-cluster-operator-64458b9458-j8gn4 1/1     Running   0           76s
PS D:\Reto 5\hexagonal-keda-strimzi\lab-keda>
```

5.- Salida del comando `kubectl get pods` mostrando brokers y zookeepers activos.

```
PS D:\Reto 5\hexagonal-keda-strimzi> kubectl apply -f lab-keda/02-kafka.yaml
kafka.kafka.strimzi.io/edu-cluster created
PS D:\Reto 5\hexagonal-keda-strimzi> kubectl get pods -n kafka -w
NAME                                READY   STATUS    RESTARTS   AGE
edu-cluster-entity-operator-fbcf96f94-svwmd  0/2     Running   0           19s
edu-cluster-kafka-0                        1/1     Running   0           49s
edu-cluster-zookeeper-0                    1/1     Running   0           118s
strimzi-cluster-operator-64458b9458-j8gn4    1/1     Running   0           9m47s
edu-cluster-entity-operator-fbcf96f94-svwmd  0/2     Running   1 (1s ago)  42s
edu-cluster-entity-operator-fbcf96f94-svwmd  0/2     Running   1 (19s ago) 60s
edu-cluster-entity-operator-fbcf96f94-svwmd  1/2     Running   1 (19s ago) 60s
edu-cluster-entity-operator-fbcf96f94-svwmd  2/2     Running   1 (44s ago) 85s
```

```
PS D:\Reto 5\hexagonal-keda-strimzi> kubectl get pods -n kafka
NAME                                READY   STATUS    RESTARTS   AGE
edu-cluster-entity-operator-fbcf96f94-svwmd  2/2     Running   1 (2m49s ago) 3m30s
edu-cluster-kafka-0                        1/1     Running   0           4m
edu-cluster-zookeeper-0                    1/1     Running   0           5m9s
strimzi-cluster-operator-64458b9458-j8gn4    1/1     Running   0           12m
PS D:\Reto 5\hexagonal-keda-strimzi>
```

```
PS D:\Reto 5\hexagonal-keda-strimzi> kubectl apply -f lab-keda/03-topic.yaml
kafkatopic.kafka.strimzi.io/scaling-demo created
PS D:\Reto 5\hexagonal-keda-strimzi> kubectl get kafkatopic -n kafka
NAME          CLUSTER    PARTITIONS  REPLICATION FACTOR  READY
scaling-demo  edu-cluster  6           1                    True
PS D:\Reto 5\hexagonal-keda-strimzi> kubectl get kafkatopic -n kafka
```

6.- Fragmento de YAML del ScaledObject, especificando los triggers de tipo Kafka y la configuración de replicas mínimas/máximas

```
05-keda.yaml x
8 spec:
14
15   minReplicaCount: 0
16   maxReplicaCount: 5
17
18   triggers:
19     - type: kafka
20       metadata:
21         bootstrapServers: edu-cluster-kafka-bootstrap.kafka.svc.cluster.local:9092
22         consumerGroup: demo-consumer-group
23         topic: scaling-demo
24         lagThreshold: "3"
25         activationLagThreshold: "1"
26         offsetResetPolicy: latest
```

7.- Screenshot de los pods de KEDA corriendo y CRDs instaladas

```
PS D:\Reto 5\hexagonal-keda-strimzi> kubectl get pods -n keda
NAME                                READY   STATUS    RESTARTS   AGE
keda-admission-webhooks-7bcfbd884d-pgblm  1/1     Running   0           3m16s
keda-operator-d676f55d7-n2mtz           1/1     Running   1 (2m46s ago)  3m16s
keda-operator-metrics-apiserver-7df995ccb4-zln5q  1/1     Running   0           3m16s
PS D:\Reto 5\hexagonal-keda-strimzi> kubectl get crd | Select-String keda

cloudeventsources.eventing.keda.sh      2026-08-15T21:39:30Z
clustercloudeventsources.eventing.keda.sh 2026-08-15T21:39:30Z
clustertriggerauthentications.keda.sh    2026-08-15T21:39:30Z
scaledjobs.keda.sh                      2026-08-15T21:39:30Z
scaledobjects.keda.sh                   2026-08-15T21:39:30Z
triggerauthentications.keda.sh           2026-08-15T21:39:30Z

PS D:\Reto 5\hexagonal-keda-strimzi> kubectl get scaledobject -n kafka
NAME                                SCALETARGETKIND  SCALETARGETNAME  MIN  MAX  READY  ACTIVE  FALLBACK  PAUSED  TRIGGERS  AUTHENTICATIONS  AGE
kafka-consumer-scaler  apps/v1.Deployment  kafka-consumer-app  0    5    True   True    False    False   kafka                                10s
PS D:\Reto 5\hexagonal-keda-strimzi> kubectl get hpa -n kafka
NAME                                REFERENCE          TARGETS  MINPODS  MAXPODS  REPLICAS  AGE
keda-hpa-kafka-consumer-scaler  Deployment/kafka-consumer-app  3/3 (avg)  1         5         2          45s
PS D:\Reto 5\hexagonal-keda-strimzi>
```

8.- Fragmento del deployment y descripción del contenedor; screenshot del pod en estado Running

```
1  apiVersion: apps/v1
2  kind: Deployment
3  metadata:
4    name: kafka-consumer-app
5    namespace: kafka
6
7  spec:
8    replicas: 0
9
10   selector:
11     matchLabels:
12       app: kafka-consumer
13
14   template:
15     metadata:
16       labels:
17         app: kafka-consumer
```

```
args:
- |
  echo "Starting Kafka consumer..."

  bin/kafka-console-consumer.sh \
    --bootstrap-server edu-cluster-kafka-bootstrap.kafka.svc.cluster.local:9092 \
    --topic scaling-demo \
    --group demo-consumer-group \
    --from-beginning \
  | while read message
  do
    echo "Processing: $message"
    sleep 2
  done
```

```
PS D:\Reto 5\hexagonal-keda-strimzi> kubectl get pods -n kafka
NAME                                READY   STATUS    RESTARTS   AGE
edu-cluster-entity-operator-fbcf96f94-swvmd   2/2     Running   1 (82m ago)  82m
edu-cluster-kafka-0                          1/1     Running   0           83m
edu-cluster-zookeeper-0                     1/1     Running   0           84m
my-producer                                0/1     Completed 0           27m
strimzi-cluster-operator-64458b9458-j8gn4     1/1     Running   0           92m
PS D:\Reto 5\hexagonal-keda-strimzi>
```

9.- Captura comparativa de réplicas antes/durante/después del escalado

```
PS D:\Reto 5\hexagonal-keda-strimzi> kubectl get scaledobject -n kafka
NAME                                SCALETARGETKIND  SCALETARGETNAME  MIN  MAX  READY  ACTIVE  FALLBACK  PAUSED  TRIGGERS  AUTHENTICATIONS  AGE
kafka-consumer-scaler              apps/v1.Deployment  kafka-consumer-app  0    5    True   False   False     False   kafka                     53m
PS D:\Reto 5\hexagonal-keda-strimzi> kubectl get hpa -n kafka
NAME                                REFERENCE                                TARGETS          MINPODS  MAXPODS  REPLICAS  AGE
keda-hpa-kafka-consumer-scaler      Deployment/kafka-consumer-app            <unknown>/3 (avg)  1         5         0          54m
PS D:\Reto 5\hexagonal-keda-strimzi>
```

```
Events:
Type      Reason              Age   From      Message
----      -
Normal    KEDAScalersStarted  50m   keda-operator   Scaler kafka is built
Normal    KEDAScalersStarted  50m   keda-operator   Started scalers watch
Normal    ScaledObjectReady   50m   keda-operator   ScaledObject is ready for scaling
Normal    ScaledObjectReady   50m   keda-operator   ScaledObject is ready for scaling
Normal    KEDAScaleTargetActivated  50m   keda-operator   Scaled apps/v1.Deployment kafka/kafka-consumer-app from 0 to 1, triggered by s0-kafka-scaling-demo
Normal    KEDAScaleTargetActivated  28m   keda-operator   Scaled apps/v1.Deployment kafka/kafka-consumer-app from 0 to 1, triggered by s0-kafka-scaling-demo
Normal    KEDAScaleTargetDeactivated  27m   keda-operator   Deactivated apps/v1.Deployment kafka/kafka-consumer-app from 2 to 0
Normal    KEDAScaleTargetActivated  25m   keda-operator   Scaled apps/v1.Deployment kafka/kafka-consumer-app from 0 to 1, triggered by s0-kafka-scaling-demo
Normal    KEDAScaleTargetDeactivated  24m   keda-operator   Deactivated apps/v1.Deployment kafka/kafka-consumer-app from 5 to 0
Normal    KEDAScaleTargetActivated  21m   keda-operator   Scaled apps/v1.Deployment kafka/kafka-consumer-app from 0 to 1, triggered by s0-kafka-scaling-demo
Normal    KEDAScaleTargetDeactivated  20m   keda-operator   Deactivated apps/v1.Deployment kafka/kafka-consumer-app from 5 to 0
PS D:\Reto 5\hexagonal-keda-strimzi>
```

Evidencia del escalamiento automático: KEDA detectó el lag generado en el topic scaling-demo y escaló dinámicamente el Deployment kafka-consumer-app desde 0 hasta un máximo de 5 réplicas. Una vez procesados los mensajes, KEDA desactivó el escalamiento y redujo nuevamente las réplicas a 0.

10. Texto de reflexión personal (5 líneas mínimo).

La práctica permitió comprender cómo diferentes tecnologías pueden integrarse para construir una solución cloud-native. La arquitectura hexagonal permite separar la lógica de negocio de componentes externos como Kafka, facilitando el mantenimiento y evolución de la aplicación. Strimzi simplifica la administración de Kafka dentro de Kubernetes mediante recursos declarativos. KEDA permite reaccionar automáticamente ante la cantidad de mensajes pendientes y escalar los consumidores según la demanda.